Take home Quiz :-

1) $\frac{dv}{dt} = v_0(t) = 2e^{-t}$

$$\frac{dN_A}{dt} = v_0 C_{A,in} - k C_A V$$

$$V \frac{dC_A}{dt} + C_A \frac{dv}{dt} = v_0 C_{A,in} - k C_A v$$

$$\frac{dv}{dt} = v_0$$

$$v \frac{dC_A}{dt} + C_A v_0 = v_0 C_A - k C_A v$$

$$\frac{dC_A}{dt} = \frac{v_0}{v}(C_{A,in} - C_A) - k C_A$$

$$\frac{dN_B}{dt} = r_B v = k C_A v$$

$$V \frac{dC_B}{dt} + C_B v_0 = k C_A v$$

$$\frac{dC_B}{dt} = -\frac{v_0}{v} C_B + k C_A$$

2)